COMPUTER GRAPHICS
PROFESSOR RI YU

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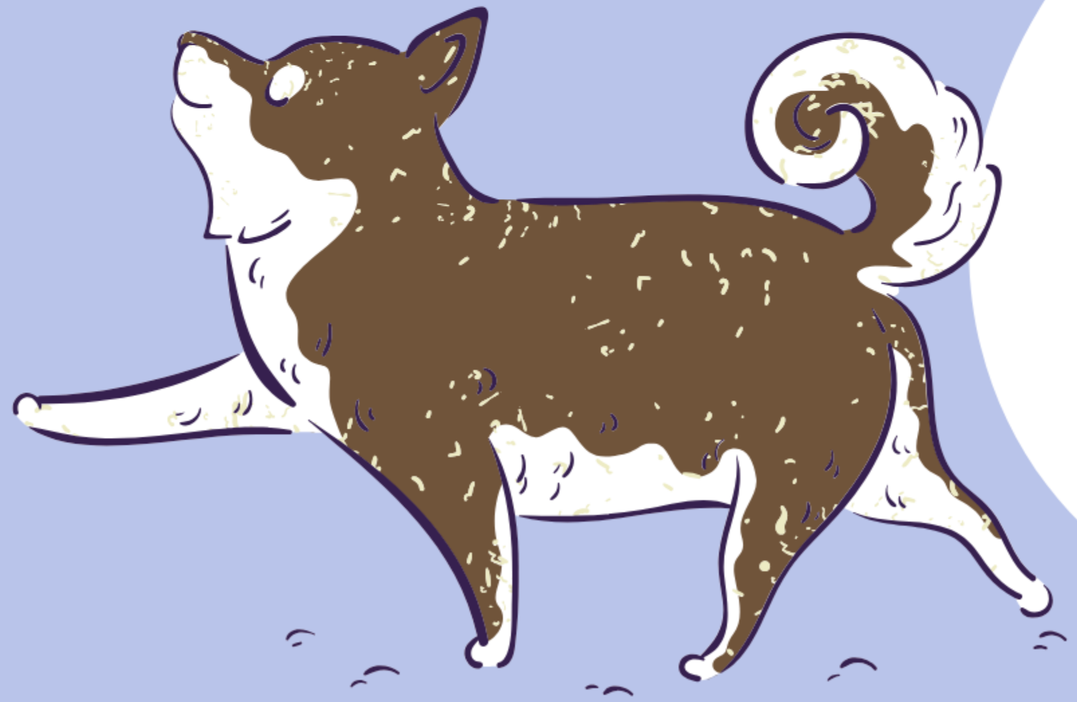
202221129 DAHEE JANG

DOG MOTION SIMULATOR & ANIMATION CREATOR



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DOG GROOMING & TRAINING

OUR GOALS

- CREATE A 3D HIERARCHICAL MODEL
- ENABLE THE DOG TO PERFORM BASIC MOTIONS
- COMBINE BASIC MOTIONS AND MODEL ANGLES TO CREATE DESIRED ANIMATIONS

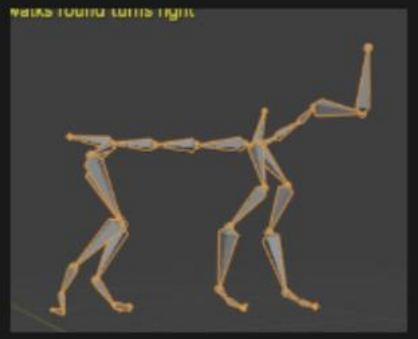
simplification process

1. 루트의 offset 0 0 0으로 변경

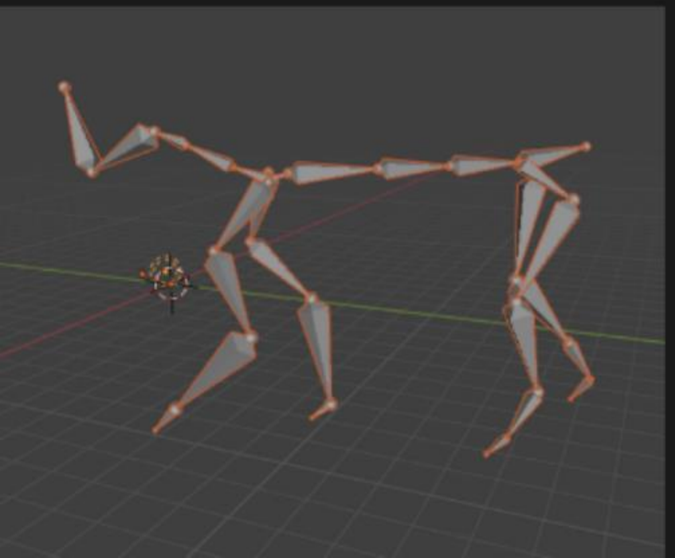
```
ROOT Storm:Solving:Root
{
    OFFSET 0.000000 0.000000 0.000000
```

2. R_Clav_1(튀어나온 부분), New Bone_4, New Bone_5, L_pow, R_Pow, L_Foot_End, R_Foot_End(ㅂ
잡한 발 부분), 해당 element 제거 (파이썬 이용)

→ 제거 전 모습



→ 제거 후 모습



음)

- Deep Motion : (bvh 파일로 다운 받을 수 있음) 이것도 사람만 가능

SayMotion: Text to 3D Animation | Open Beta Now Available! (youtube.com)

Deepmotion 및 Blender를 사용한 자동 모션 캡처 (garagefarm.net)

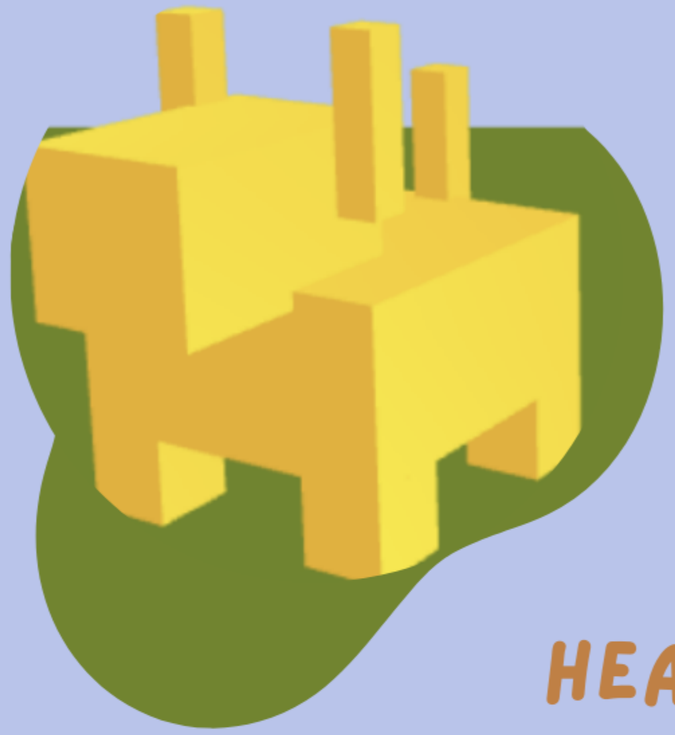
직접 영상 녹화 한 후 Deep Motion으로 실행해 본 결과 잘 됨 (춤 영상)

├→해당 코드 (bvh파일)

TRIALS

2024-webgl-dog-simulation / **webgl_dog_simulation**

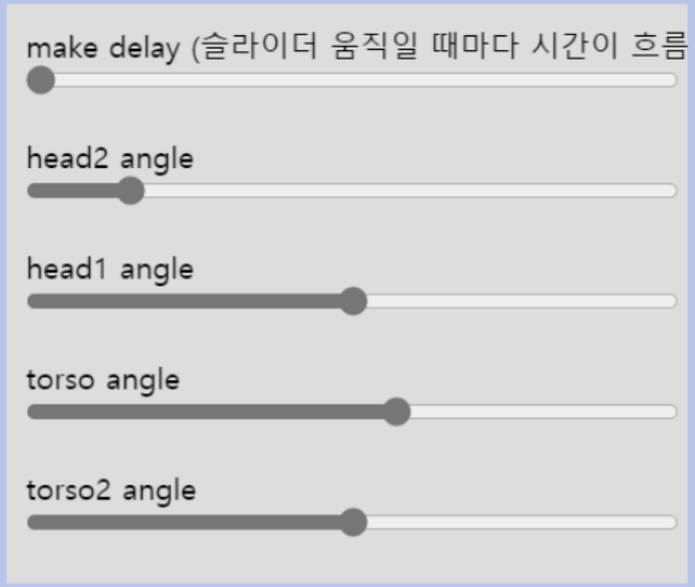
- HumanSkeleton
- OurSkeleton
- bvh
- fbx
- original_bvh
- bvh2webgl.ipynb
- dahee.html
- dahee.js

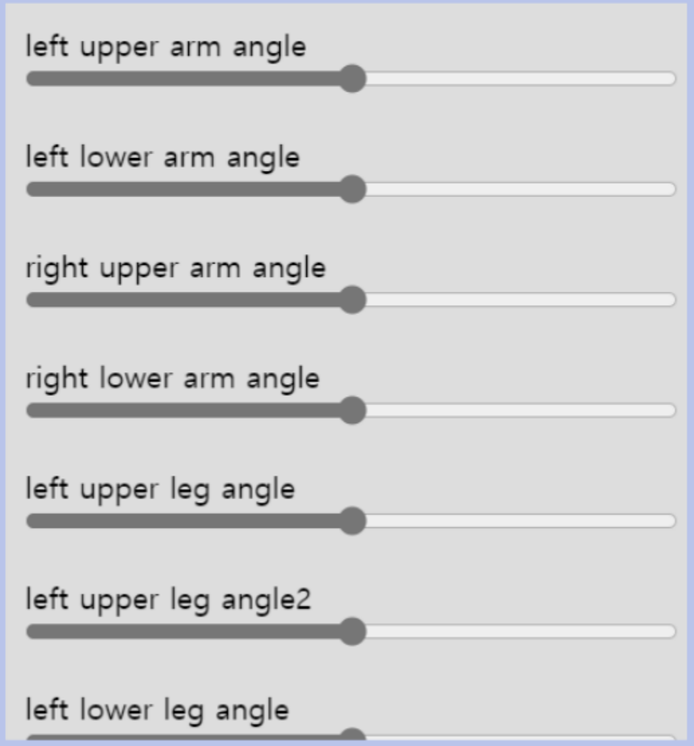


MAIN MODEL

- TORSO
  - HEAD
    - LEFT EAR
    - RIGHT EAR
  - LEFT UPPER ARM
    - LEFT LOWER ARM
  - RIGHT UPPER ARM
    - RIGHT LOWER ARM
  - LEFT UPPER LEG
    - LEFT LOWER LEG
  - RIGHT UPPER LEG
    - RIGHT LOWER LEG
  - TAIL

SLIDERS





WE CAN CHANGE ANGLE

BASIC MOTIONS

WALK

RUN

LIE DOWN

Walk

Run

Lie Down

Eat

Pee

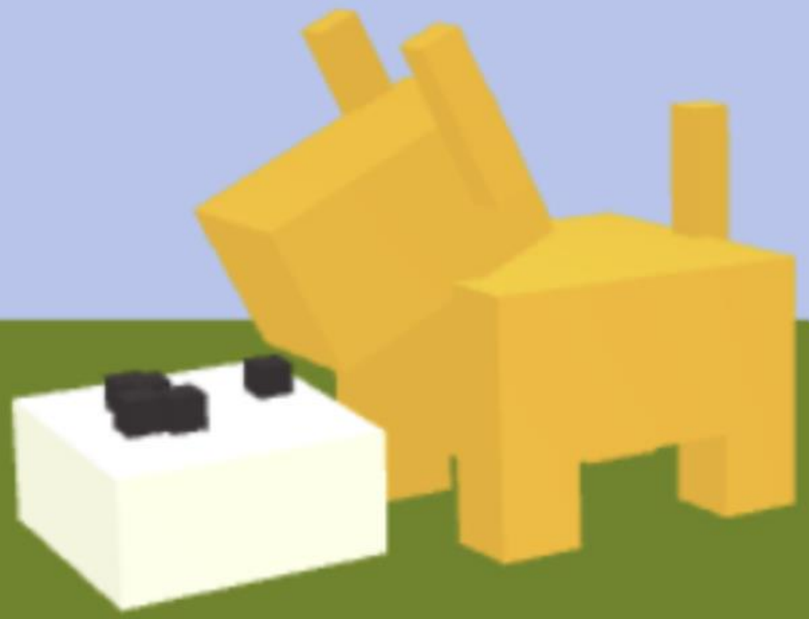
Shake

Wag Tail

Stop

BASIC MOTIONS

EAT



PEE

WAG TAIL

ANIMATION CREATOR

01 Press the "motioncapture start" button to begin recording the motion capture.

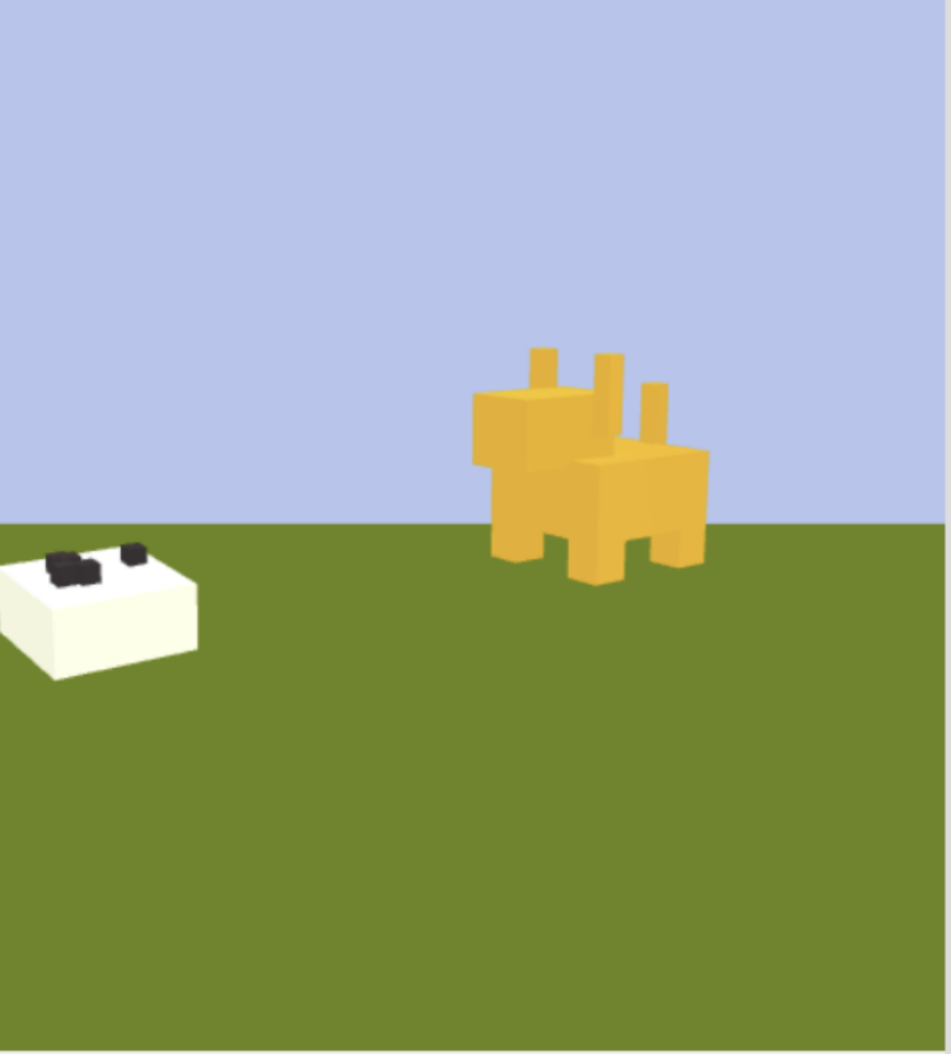
02 Use the sliders or buttons to create movements for the dog.

03 Press the "motioncapture stop" button to stop the recording.

04 Press the "motioncapture play" button to replay the recorded motion.

05 Press the "motioncapture print" button to output the recorded motion data to the console for use.

06 Use motion data to save animation and create new button



Press the "motioncapture start" button to begin recording the
otion capture.
Use the sliders or buttons to create movements for the dog.
Press the "motioncapture stop" button to stop the recording.
Press the "motioncapture play" button to replay the recorded
otion.
Press the "motioncapture print" button to output the
corded motion data to the console for use.

make delay (슬라이더 움직일 때마다 시간이 흐름)

head2 angle

head1 angle

torso angle

torso2 angle

left upper arm angle

left lower arm angle

right upper arm angle

right lower arm angle

left upper leg angle

left upper leg angle2

left lower leg angle

record the motion capture

print the recorded motion data

replay the recorded motion

motioncapture start

motioncapture stop

reset all

motioncapture print

motioncapture play

Walk

Run

Lie Down

Eat

Pee

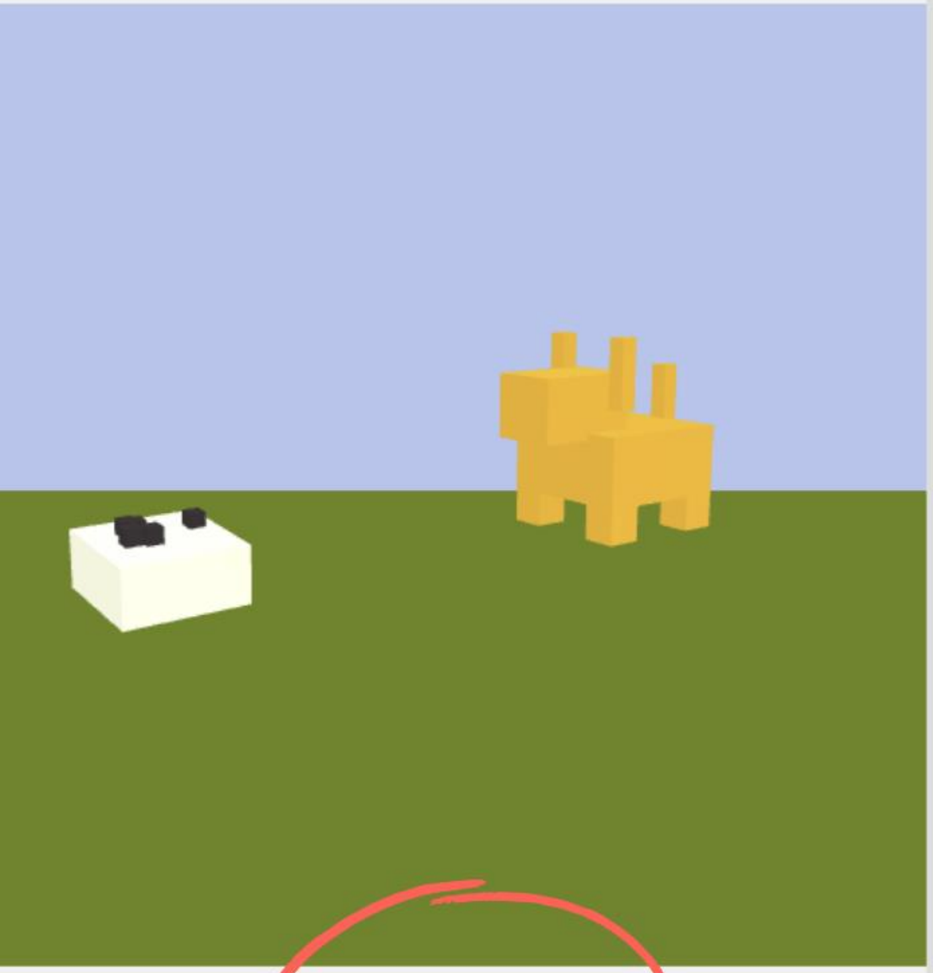
Shake

Wag Tail

Stop

Return to main pag

YOU CAN CREATE BUTTONS TO SAVE YOUR ANIMATIONS ON THIS PAGE!



Make animation yourself.

reset all

Play with me, Master!

Hand, please.

Not this, Master.

Pee.

Eat.

Play with me! Master!!!

Run away.

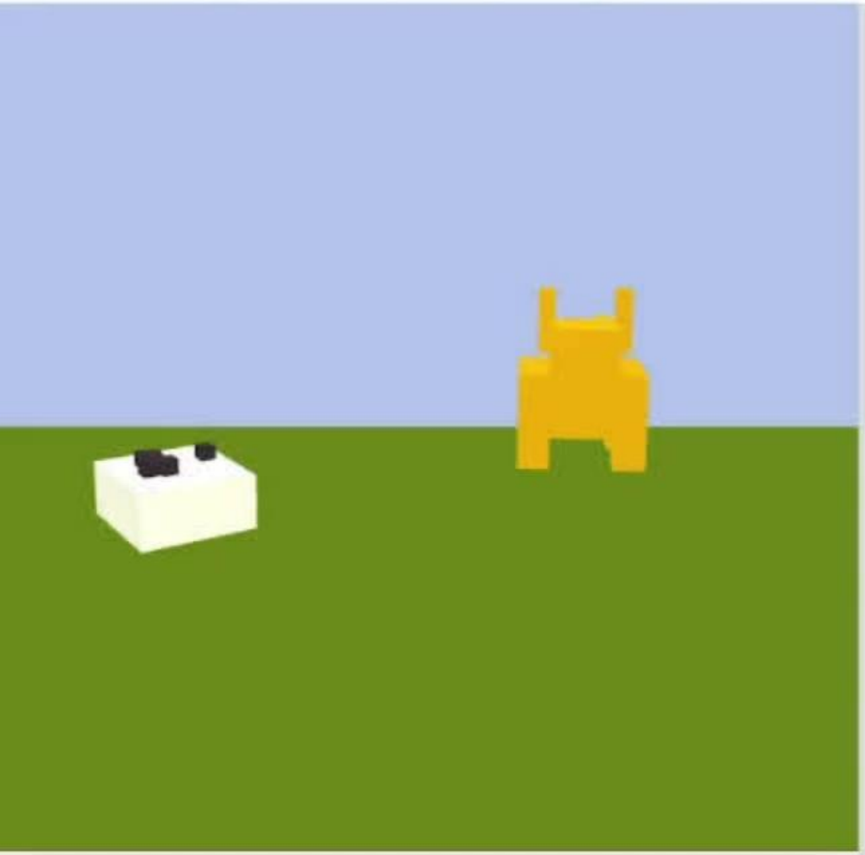
Walk away.

Did you call me?

ANIMATIONS WE MADE

IF YOU CLICK,
YOU CAN MAKE
ANIMATION IN PERSON

WITH BASIC MOTIONS
AND ADJUSTING ANGLE!



make delay (슬라이더 움직일 때마다 시간이 흐름)

head2 angle

head1 angle

torso angle

torso2 angle

left upper arm angle

left lower arm angle

right upper arm angle

right lower arm angle

left upper leg angle

left upper leg angle2

left lower leg angle

motioncapture start

motioncapture stop

reset all

motioncapture print

motioncapture play

Walk

Run

Lie Down

Eat

Pee

Shake

Wag Tail

Stop

Return to main page

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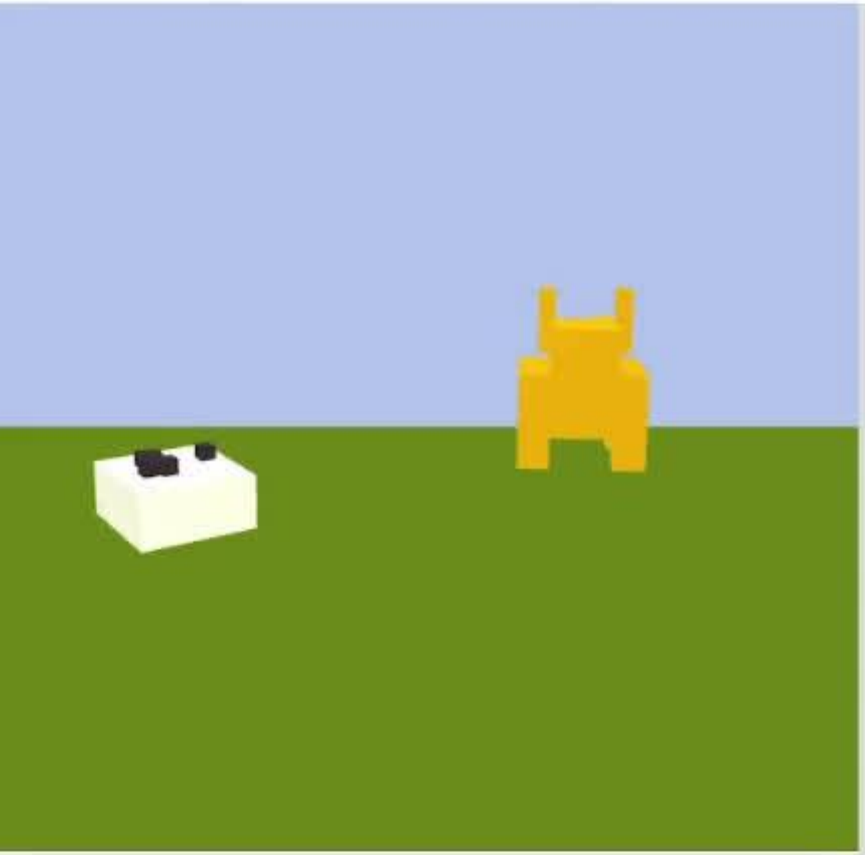
MOTION CAPTURE

Sublet-K/ItHome-Front

127.0.0.1:5500/motioncapture

127.0.0.1:5500/motioncapture.html



make delay (슬라이더 움직일 때마다 시간이 흐름)

head2 angle

head1 angle

torso angle

torso2 angle

left upper arm angle

left lower arm angle

right upper arm angle

right lower arm angle

left upper leg angle

left upper leg angle2

left lower leg angle

motioncapture start

motioncapture stop

reset all

motioncapture print

motioncapture play

Walk

Run

Lie Down

Eat

Pee

Shake

Wag Tail

Stop

Return to main page.

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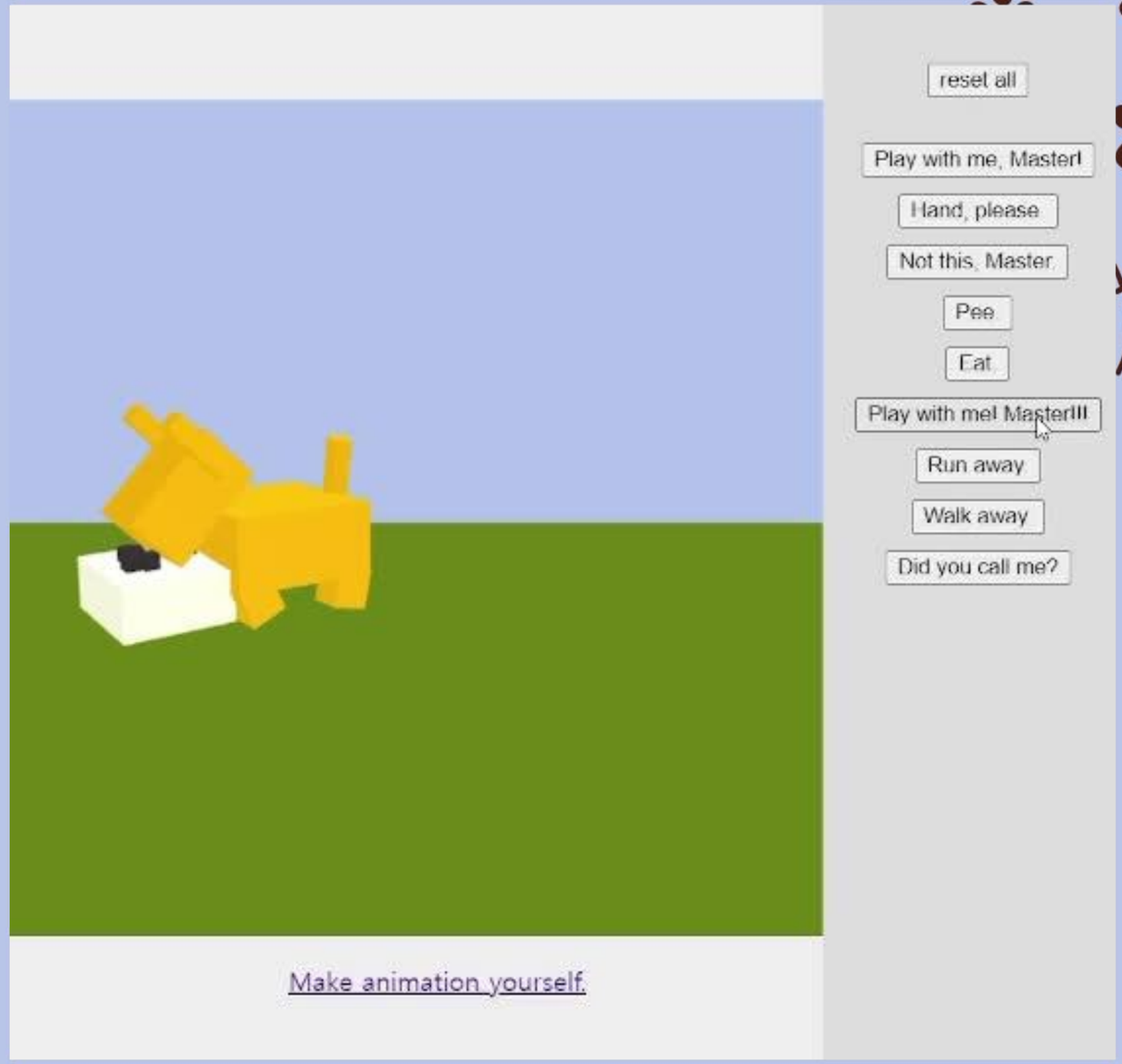
ANIMATIONS

WE MADE

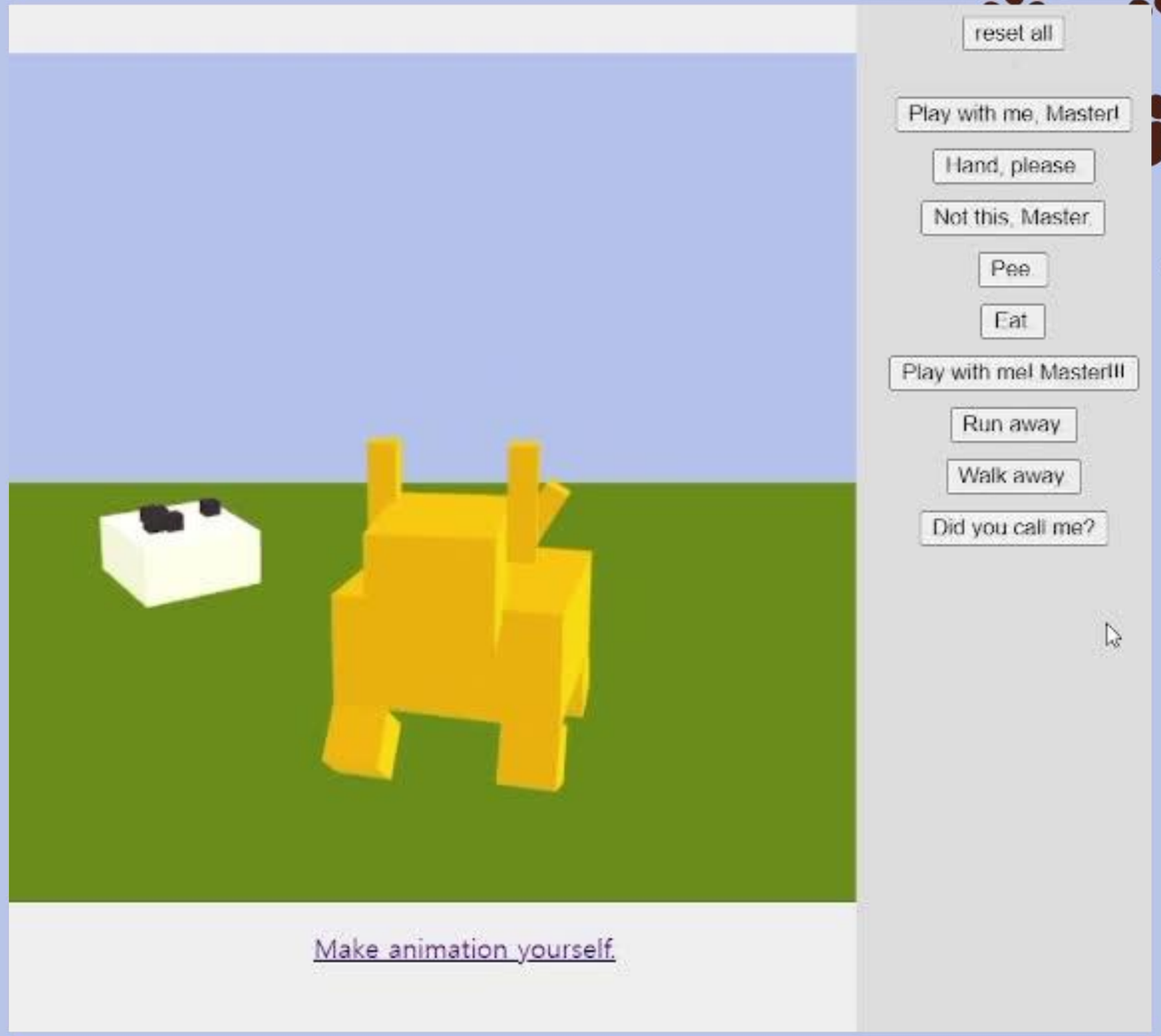
EAT.



ANIMATIONS

WE MADE

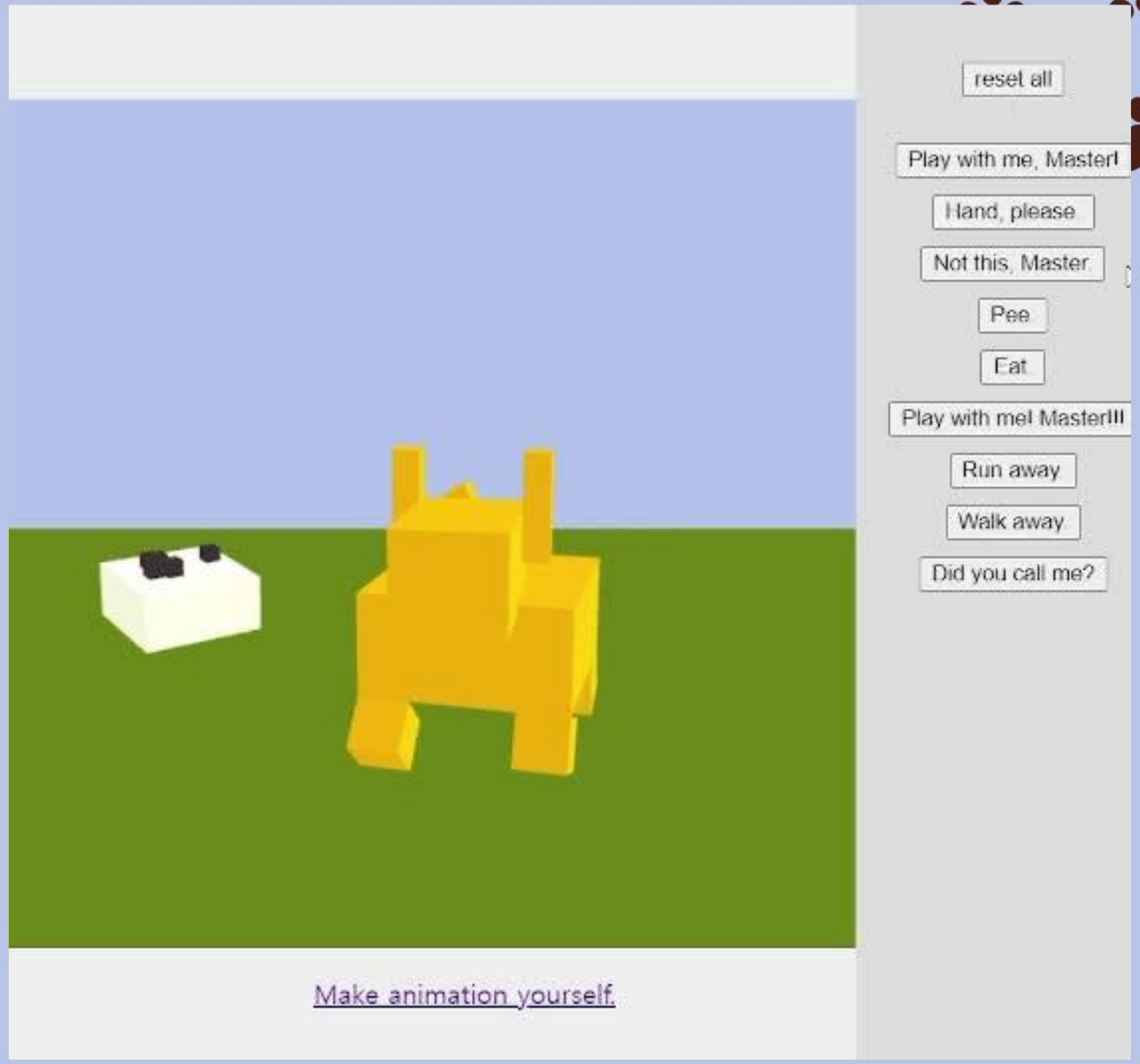
DID YOU
CALL ME?



ANIMATIONS

WE MADE

HAND, PLEASE



ROLE

	MODELING	BASIC MOTIONS	MOTION CAPTURE	CREATE ANIMATIONS	PPT	REPORT
LEADER	DAHEE	DAHEE	MINJAE	MINJAE	DAHEE	MINJAE, DAHEE
ASSISTANT	-	MINJAE	-	DAHEE	MINJAE	-

THANK YOU